Name: ______________________________ Phys 12

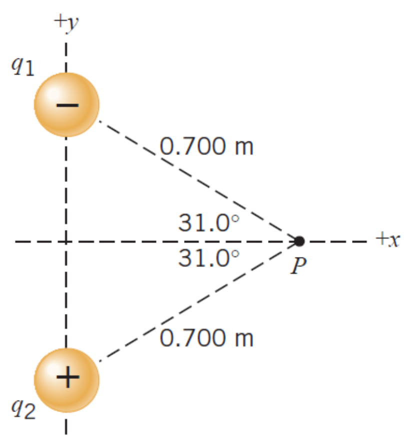
10) What is the electric field strength at a point in space where an electron experiences an initial acceleration of 7.50×10^{12} m/s^2? (42.7 N/C)

11) The electric field strength at a distance of 3.00×10^{-1} m from a charged object is 3.60×10^5 N/C. What is the electric field strength at a distance of 45 cm from the same object? (1.60×10^5 N/C)

12) An electric field of 260 000 N/C points due west at a certain spot. What are the magnitude and direction of the force that acts on a charge of – 7.0 µC at this spot? (1.8 N due east)

13) Two charges, -16 µC and + 4.0 µC, are fixed in place and separated by 3.0 m. a) At what spot along a line through the charges is net electric field zero? Locate this spot relative to the positive charge. *(Hint: the spot does not necessarily lie between the two charges.)* b) What would by the force on a charge of + 14 µC placed at this spot? (3.0 m from the positive charge (not between the charges), 0N)

14) Two point charges are lying on the y axis where q_1 = - 4.00 µC and q_2 = + 4.00 µC. They are equidistant from the point P, which lies on the x axis. a) What is the net electric field at P? b) A small object of charge q_o = +8.00 µC and mass m = 1.20 g is placed at P. When it is released, what is its acceleration?
(7.56×10^4 N/C, directed along the +y axis, 5.04×10^2 m/s^2, along the +y axis)



*15) At three corners of a rectangle (length = 2d, height = d), the following charges are located: $+q_1$ (upper left corner), $+q_2$ (lower right corner), and -q (lower left corner). The net electric field at the (empty) upper right corner is zero. Find the magnitudes of q_1 and q_2. Express your answers in terms of q.
($|q_1| = 0.716\,q$, $|q_2| = 0.0895q$)